

Get started with the course

Intro to Course 4

Econometrics

- A branch of economics that uses statistics to analyze economic problems

What you will learn

- The role of stats in data analytics
- Descriptive statistics
- Inferential statistics
- Probability
- Sampling
- Confidence intervals
- Hypothesis testing

Tips for starting a career in data analytics

- Be able to present your results
- Master the basics, and don't try to go too fast
- Meet other people in the field

Welcome to module 1

- Fundamental stats concepts
- A/B test
- Descriptive statistics
- Inferential statistics

- Types of descriptive statistics
- Python

The role of statistics in data science

Data professionals use statistical methods to

- Identify meaningful patterns and relationships in data
- Analyze and quantify uncertainty
- Generate insights from data
- Make informed predictions about the future
- Solve complex problems

Statistical concepts

- Probability
- Average
- Margin of error
- Percentile
- Median

A data professional might use probability to

- Predict the future rate of return on an investment
- Estimate the annual average sales revenue for a company
- Calculate the margin of error to quantify the uncertainty of an employee satisfaction survey
- Use percentiles to rank median home prices in different cities

Methods

- Hypothesis testing
- Classification

- Regression
- Time series analysis

Statistics in action: A/B testing

A/B testing

- is a way to compare two versions of something to find out which version performs better
- A/B testing has become popular because it works well for many online applications.

For example, businesses often use A/B testing to create two versions of a webpage to find out which one gets more clicks, purchases, or subscriptions. Even small changes to a webpage, like changing the color, size, or location of a button can increase financial gains. A/B test help business leaders optimized product performance and improve customer experience.

You want to run an A/B test to find out if changing the size of the add to cart button will increase the conversion rate, or the percentage of customers who purchase a product. The test presents two versions of your webpage, known as version A and version B, to a group of randomly selected users.

Version A is the original webpage. Version B is the webpage with the larger add to cart button. The test directs half the users to version A and half to version B. The test runs for two weeks. When the test is over, a statistical analysis of the results indicates that the larger button in version B resulted in an increase in purchases. The conversion rate for version B is 30 percent. This is three times greater than the conversion rate for version A, which is 10 percent.

That's a notable increase. Because of your A/B test, your company has a data-driven reason for replacing the current webpage with version B and increasing the size of an add to cart button.

A/B Test Steps

- Analyzes a small group of users
- Decide on the sample size
- Determine the statistical significance

Confidence interval

- A range of values that describes the uncertainty surrounding an estimate

Descriptive statistics versus inferential statistics

Descriptive statistics describe or summarize the main features of a dataset. Descriptive stats are very useful because they let you quickly understand a large amount of data.

For example, let's say you had data on the heights of 10 million people. You probably don't want to scan 10 million rows of data to analyze it for your report.

Even if you did, it would be difficult to interpret the data. However, if you summarize the data, you can instantly make it meaningful. Finding the mean or average height gives you useful knowledge about the data plus reading a summary is much better than staring at millions of rows of data

There are two common forms of descriptive statistics, visuals like graphs and tables, and summary stats. Previously you learned how graphs and tables can help you explore, visualize, and share your data. You're likely familiar with data visualizations such as histograms, scatter plots, and box plots.

Summary statistics let you summarize your data using a single number.

A common example is the mean or average value. There are two main types of summary stats, measures of central tendency and measures of dispersion. Measures of central tendency like the mean, let you describe the center of your dataset, measures of dispersion like standard deviation, let you describe the spread of your dataset or the amount of variation in your data points.

Stats like mean and standard deviation are used to describe and summarize data. But data professionals do more than just described their data. They also draw

conclusions and make predictions based on data. For this, they use inferential statistics.

Inferential statistics

Inferential statistics allow data professionals to make inferences about a dataset based on a sample of the data. The dataset that the sample is drawn from is called the population. The population includes every possible element that you are interested in measuring. As we've discussed, a sample is a subset of a population. Data professionals use samples to make inferences about populations. In other words, they use the data they collect from a small part of the population to draw conclusions about the population as a whole.

Note that a statistical population may refer to people, objects, or events.

For instance, a population might be the set of all residents of a country, all the planets in our solar system, or all the outcomes of 1,000 coin flips. A sample is a smaller group or subset of any of these populations.

Let's check out an example. Say you want to research the music preferences of every college student in the United States to find out whether they prefer pop, rap, country, classical, or another genre of music. There are around 20 million college students in the United States and would be too expensive and time-consuming to gather data from every single person. Instead, you can use a sample and survey only a subset of the 20 million students.

Later on, we'll discuss the factors that go into choosing different sample sizes and how larger sample sizes affect your results. For now, let's imagine you decide to survey 1,000 students instead of 20 million.

Then you can use the results to make inferences about the music preferences of all college students. Keep in mind that your sample should be representative of your population. Otherwise, the conclusions you draw from your sample will be unreliable and possibly biased. A representative sample is a sample that accurately reflects the population, for example, if you only survey math majors or only student athletes, your sample will not be representative of all college students.

Finally, let's review two terms that correspond to population and sample, parameter and statistic.

A parameter is a characteristic of a population. A statistic is a characteristic of a sample, for example, the average height of the entire population of giraffes is a parameter. The average height of a random sample of 100 giraffes is a statistic. As I mentioned, it's difficult to collect data about every member of a large population. In this case, to locate and measure the height of every single giraffe in the world. We use the known value of a sample statistic, the average of 100 giraffes, to estimate the unknown value of a population parameter. That's all for now.